Exact star formation under arbitrary interfaces

Research note for discussion, 15 October 2026

For a finite simple graph, a set S is *2-star-forming* if every $x \notin S$ lies in a copy of $K_{1,2}$ in $G[S \cup \{x\}]$. Copies need not be induced. Write $\mathrm{SF}_2(G)$ for the largest size of an inclusion-minimal such set, and $\beta_2(G)$ for the largest size of a set inducing maximum degree at most one. If $d_S(x) = |N(x) \cap S|$, the exact local star test is

$$d_S(x) \geq 2 \quad \text{or} \quad \exists u \in N(x) \cap S : \ d_S(u) \geq 1. \tag{1}$$

The alternatives make x the centre or a leaf. Star formation is upward closed, so inclusion-minimality is equivalent to failure of all single deletions. The tested failure vertex may differ from the deleted vertex.

The fixed graph. Let H have parts $A = \{a_1, \ldots, a_5\}$ and $B = \{b_1, \ldots, b_5\}$, and cross-edge matrix

$$\begin{pmatrix} 1 & 1 & 0 & 0 & 1 \\ 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 & 1 \end{pmatrix}.$$

Degrees on both sides are $(3, 3, 2, 4, 4)$. Put $S_0 = \{a_1, a_2, a_3, b_1, b_2, b_3\}$. Each outside vertex has two neighbours in S_0. The following single deletions fail (1).

Deleted	a_1	a_2	a_3	b_1	b_2
Failure	b_5	b_4	a_3	a_5	a_4
Remaining neighbours	$\{a_3\}$	$\{a_3\}$	$\varnothing$	$\{b_3\}$	$\{b_3\}$

Deleting b_3 fails at b_3, with no remaining selected neighbour. Vertices a_3, b_3 are isolated in $H[S_0]$, so the singleton neighbours in the table cannot act as centres. Thus S_0 is minimal and $\mathrm{SF}_2(H) \geq 6$.

Dissociation bound. The matrix is symmetric under exchanging a_i, b_i. A dissociation six-set has side split $(1, 5)$, $(2, 4)$ or $(3, 3)$, up to this symmetry. The first split violates minimum degree two. In $(2, 4)$, omitting one B vertex can lower selected degree to at most one only for a_3, so two selected A vertices are impossible. In $(3, 3)$, the degree-four vertices a_4, a_5 are excluded, forcing a_1, a_2, a_3. Both a_1, a_2 then need their common pair b_1, b_2 omitted, leaving both b_4, b_5 adjacent to a_3. Thus $\beta_2(H) \leq 5$; the independent set A attains five.

Theorem 1. *Let I be any finite set and $R \subseteq I \times I$. Take disjoint copies H_i and add $b_1^{(i)} a_1^{(j)}$ for every $(i, j) \in R$, obtaining F_R. Then $\beta_2(F_R) = 5|I|$ and $\mathrm{SF}_2(F_R) = 6|I|$. For a directed cycle on $r \geq 2$ indices, F_R is two-connected and bipartite, has maximum degree four, order $10r$, and gap exactly r.*

The upper bound, including $\mathrm{SF}_2(H) \leq 6$, follows from the local argument on the next two pages. No enumeration is needed for these assertions.

A dense selection admits a safe internal deletion. Only the ports $p = a_1, q = b_1$ may meet vertices outside a copy of H.

Lemma 1. *For $T \subseteq V(H)$ with $|T| \geq 6$, star formation is equivalent to domination: every vertex outside T has a neighbour in T.*

Proof. Star formation implies domination. Conversely, suppose $x \notin T$ has the unique selected neighbour u, and u has no neighbour in T. Then both $N(x) \setminus \{u\}$ and $N(u)$ lie in $Q = V(H) \setminus T$. They are disjoint by bipartiteness. Every edge xu satisfies $d(x) + d(u) \geq 6$: the degree-two vertices have only degree-four neighbours, and other endpoints have degree at least three. Consequently

$$|Q| \geq d(x) - 1 + d(u) \geq 5,$$

contradicting $|Q| \leq 4$. Thus every singleton neighbour can serve as a centre; vertices with two selected neighbours are themselves centres. □

The closed neighbourhoods of size at most four are precisely

$$\begin{aligned} Z_{a_3} &= \{a_3, b_4, b_5\}, & Z_{b_3} &= \{b_3, a_4, a_5\}, \\ Z_{a_1} &= \{a_1, b_1, b_2, b_5\}, & Z_{a_2} &= \{a_2, b_1, b_2, b_4\}, \\ Z_{b_1} &= \{b_1, a_1, a_2, a_5\}, & Z_{b_2} &= \{b_2, a_1, a_2, a_4\}. \end{aligned}$$

The two triples are disjoint, and any two distinct displayed quadruples intersect in at most two vertices. All other closed neighbourhoods have size five.

Lemma 2. *If $U \subseteq V(H)$ dominates H and $|U| \geq 7$, there is $v \in U \setminus \{p, q\}$ such that $U \setminus \{v\}$ is star-forming inside H, and each selected port having a selected internal neighbour before deletion still has one afterwards.*

Proof. Put $W = V(H) \setminus U$, so $|W| \leq 3$. A candidate deletion v destroys domination exactly when some $N[x]$ is contained in $W \cup \{v\}$. Since U dominates, v must be the unique element of $N[x] \setminus W$.

At most one candidate completes one of the two triples: each requires two members of that triple already in W, whereas the triples are disjoint. At most one candidate completes a quadruple: this requires $|W| = 3$ and W contained in that quadruple, and two distinct quadruples cannot both contain the same three-element W. A size-five neighbourhood is too large. At most two candidates are therefore bad for domination.

For each port, at most one further candidate destroys its internal selected-neighbour activity, namely its unique neighbour in U, if it has one. Thus at most four candidates are bad in total. There are at least $|U| - 2 \geq 5$ candidates outside the ports, so a safe candidate remains. Its deletion preserves domination and leaves at least six vertices; Lemma 1 gives star formation. It also preserves port activity as required. □

The count is a union bound on bad deletion choices. Overlaps are harmless; no surplus or witness is assigned twice to justify different deletions.

The local bound in an arbitrary surrounding graph.

Lemma 3. *If G contains an induced H whose external edges touch only p, q, then every inclusion-minimal 2-star-forming set S of G satisfies $|S \cap V(H)| \leq 6$.*

Proof. Suppose $U = S \cap V(H)$ has size at least seven. First U dominates H. Otherwise some $x \notin U$ has $N[x] \subseteq V(H) \setminus U$, a set of size at most three. The list of closed neighbourhoods forces $x = a_3$ or b_3. These are not ports, so x has no selected neighbour anywhere in G, contradicting star formation.

Choose v from Lemma 2 and set $S' = S \setminus \{v\}$. Every vertex of the copy outside S' has a star entirely inside the copy. For a vertex x outside the copy and outside S', take its old witnessing star. If that star avoids v, it survives. Otherwise x is not adjacent to the non-port v, so v must be a leaf, the centre must be a selected port u, and the other leaf must be x. The port had the selected internal neighbour v, so after deletion it has some selected internal neighbour w. The star with centre u and leaves x, w survives. Thus S' is star-forming, contradicting minimality. □

Proof of the theorem. Applying Lemma 3 to each copy bounds any minimal set by $6|I|$. The union of the sets S_0 attains this size and remains minimal: the outside vertices keep their two selected neighbours, and all six deletion witnesses from page 1 persist. Interfaces only meet a_1, b_1, so the possible singleton centres a_3, b_3 remain isolated. Upward closure then proves minimality. This gives $\mathrm{SF}_2(F_R) = 6|I|$.

Every dissociation set restricts to at most five vertices in each induced copy. The full A part attains $5|I|$, since every interface joins B to A. Hence $\beta_2(F_R) = 5|I|$. Loops of R only specify the already existing edge b_1a_1 and do not affect the argument.

Each copy contains the Hamilton cycle

$$a_1, b_1, a_2, b_2, a_4, b_3, a_5, b_4, a_3, b_5, a_1.$$

In the cyclic family, deleting a vertex leaves its copy connected and destroys at most one interface, because the two ports are distinct. The remaining interfaces still connect all copies. For $r = 2$ at least one of the two distinct interfaces remains. Thus the graph is two-connected. The ports rise from degree three to four and all other degrees stay fixed. □

Verification and scope. Independent base enumeration returns $\beta_2(H) = 5$, $\mathrm{SF}_2(H) = 6$ and 37, 48, 1 minimal sets of sizes 4, 5, 6. Separate controls check all 174 dominating selections of size at least seven and 29,656 safe global deletions in 60 surrounding graphs. These finite controls check the argument; the all-size conclusion follows from the proofs above. The accompanying Lean sources check the literal/local definition, witness preservation and the general local-deletion argument; the precise completed declarations are listed with the source release. This note does not settle a sharp universal gap coefficient, computational complexity, or the rectangular-grid triple equality.